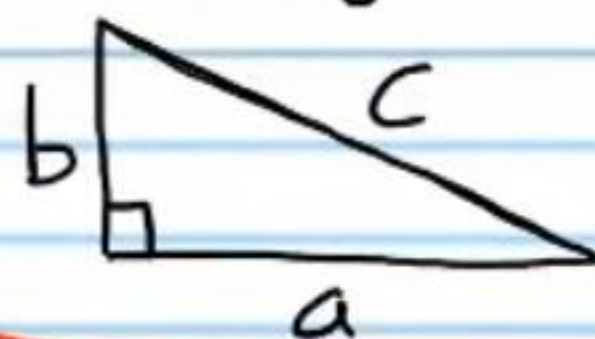


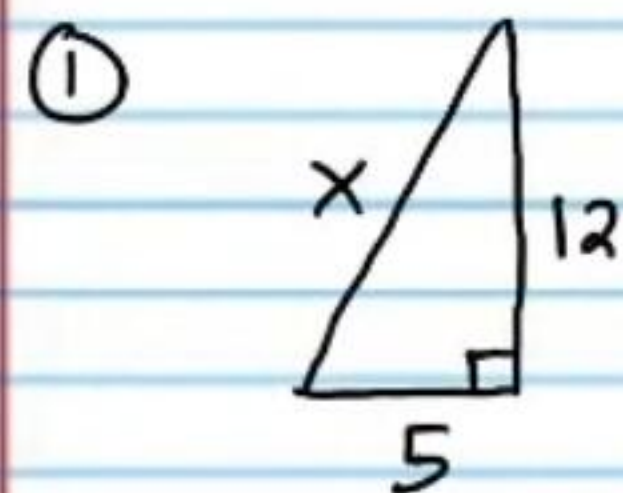
Right Triangles (Part I): Geometric Mean Theorems, Special Right Triangles, and Pythagorean Theorems

Review on the Pythagorean Theorem

The Pythagorean Theorem: If a , b , and c are the legs and hypotenuse of a right triangle, respectively, then $a^2 + b^2 = c^2$.



Find the unknown values.



$$a^2 + b^2 = c^2$$

$$5^2 + 12^2 = x^2$$

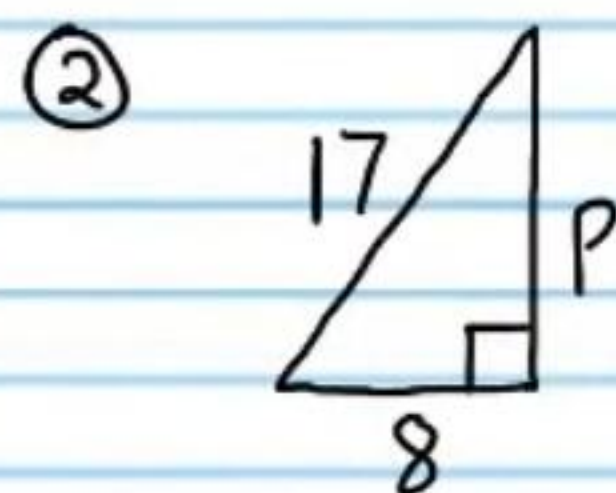
$$25 + 144 = x^2$$

$$169 = x^2$$

$$\pm\sqrt{169} = \sqrt{x^2}$$

$$\pm 13 = x$$

$$\boxed{13 = x}$$



$$a^2 + b^2 = c^2$$

$$8^2 + p^2 = 17^2$$

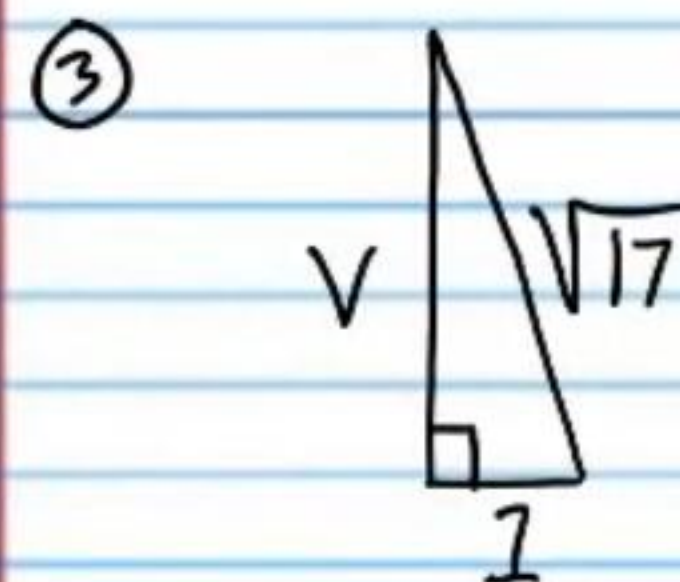
$$64 + p^2 = 289$$

$$p^2 = 225$$

$$\sqrt{p^2} = \pm\sqrt{225}$$

$$p = \pm 15$$

$$\boxed{p = 15}$$



$$a^2 + b^2 = c^2$$

$$v^2 + 1^2 = (\sqrt{17})^2$$

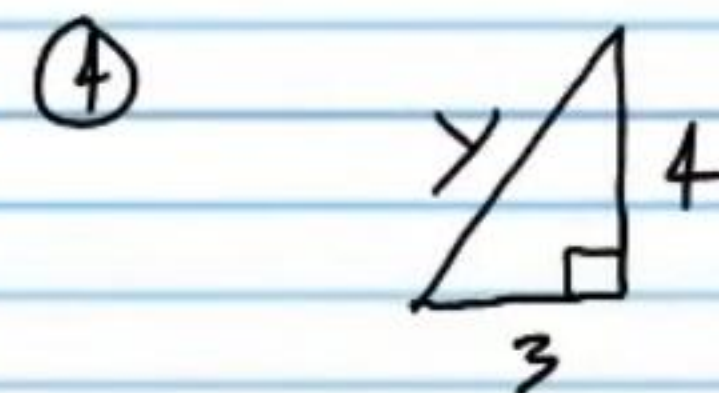
$$v^2 + 1 = 17$$

$$v^2 = 16$$

$$\sqrt{v^2} = \pm\sqrt{16}$$

$$v = \pm 4$$

$$\boxed{v = 4}$$



$$3^2 + 4^2 = y^2$$

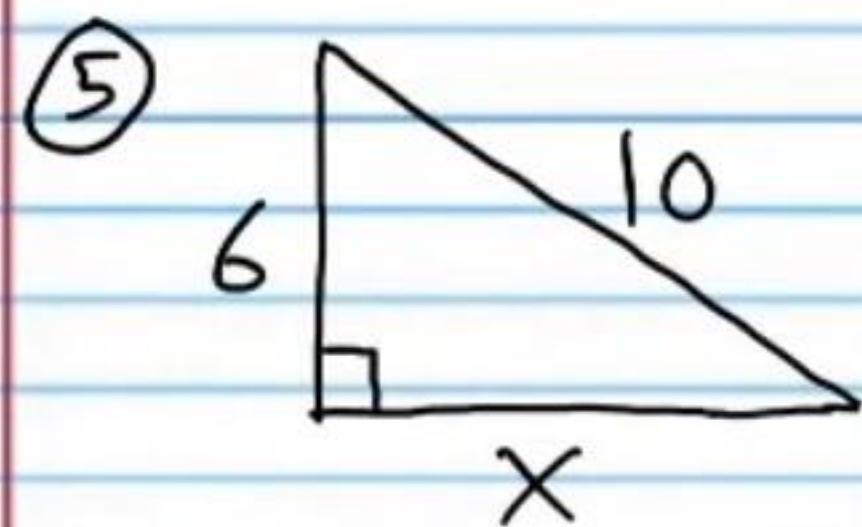
$$9 + 16 = y^2$$

$$25 = y^2$$

$$\pm\sqrt{25} = \sqrt{y^2}$$

$$\pm 5 = y$$

$$\boxed{5 = y}$$



$$6^2 + x^2 = 10^2$$

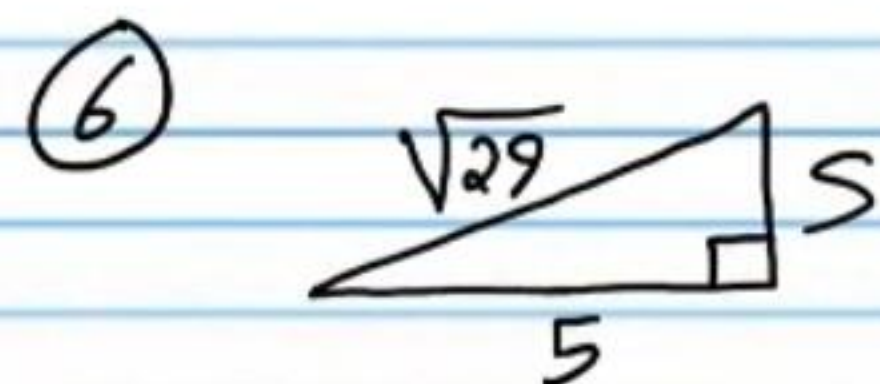
$$36 + x^2 = 100$$

$$x^2 = 64$$

$$\sqrt{x^2} = \pm \sqrt{64}$$

$$x = \pm 8$$

$$\boxed{x = 8}$$



$$5^2 + s^2 = (\sqrt{29})^2$$

$$25 + s^2 = 29$$

$$s^2 = 4$$

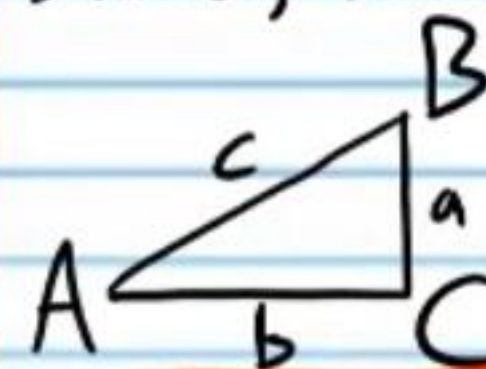
$$\sqrt{s^2} = \pm \sqrt{4}$$

$$s = \pm 2$$

$$\boxed{s = 2}$$

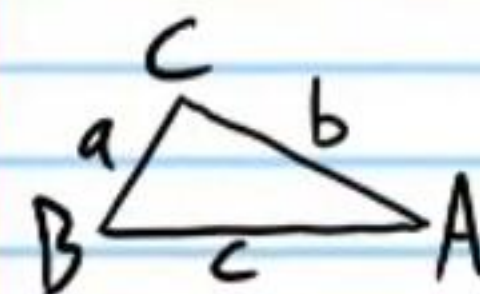
The Converse of the Pythagorean Theorem

Pythagorean Converse Theorem: If the square of the length of the longest side of a triangle is equal to the sum of the squares of the lengths of the other sides, then the triangle is a right triangle.



If $c^2 = a^2 + b^2$, then $\triangle ABC$ is a right triangle.

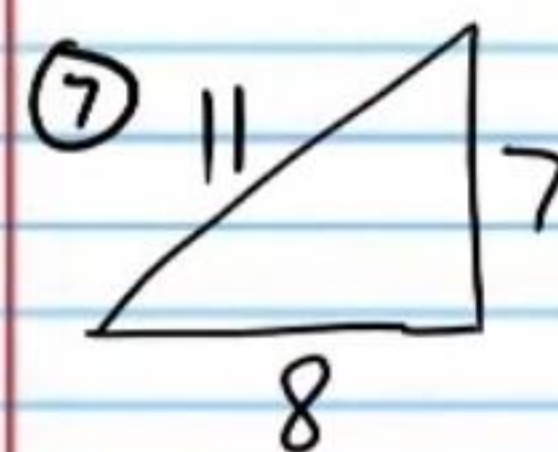
Pythagorean Inequalities Theorem: If the square of the length of the longest side of a triangle is less than the sum of the squares of the lengths of the other sides, then the triangle is an acute triangle. But if the square of the length of the longest side is greater than the sum of the squares of the lengths of the other sides, then the triangle is an obtuse triangle.



I) If $c^2 < a^2 + b^2$, then $\triangle ABC$ is acute.

II) If $c^2 > a^2 + b^2$, then $\triangle ABC$ is obtuse.

Are the following triangles acute, obtuse, or right triangles?



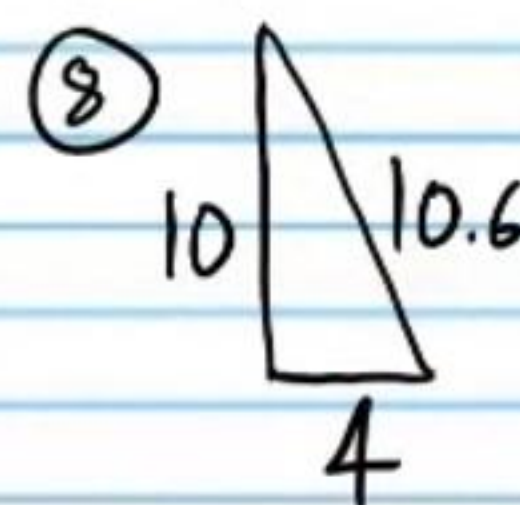
$$c^2 \bigcirc a^2 + b^2$$

$$11^2 \bigcirc 8^2 + 7^2$$

$$121 \bigcirc 64 + 49$$

$$121 > 113$$

Obtuse



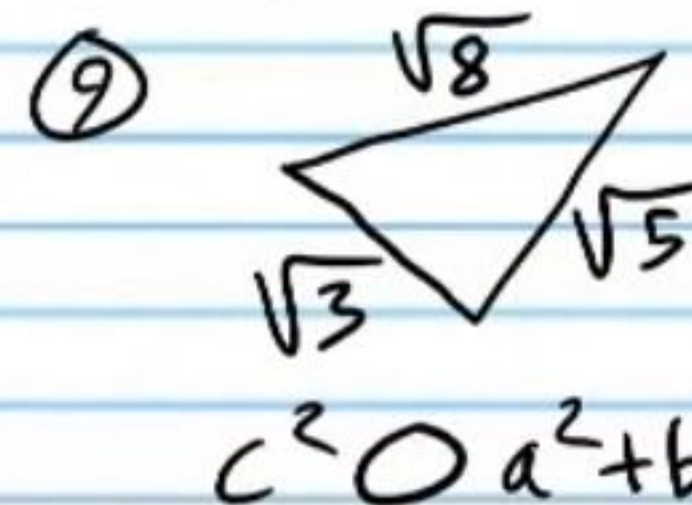
$$c^2 \bigcirc a^2 + b^2$$

$$(10.6)^2 \bigcirc 4^2 + 10^2$$

$$112.36 \bigcirc 16 + 100$$

$$112.36 < 116$$

Acute



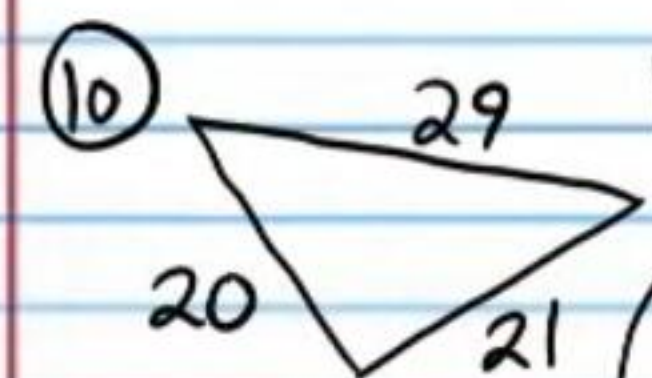
$$c^2 \bigcirc a^2 + b^2$$

$$(\sqrt{8})^2 \bigcirc (\sqrt{3})^2 + (\sqrt{5})^2$$

$$8 \bigcirc 3 + 5$$

$$8 \equiv 8$$

Right



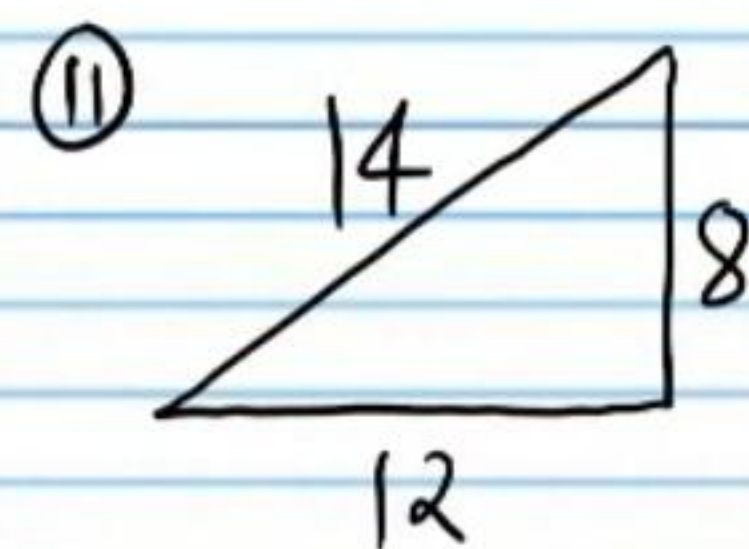
$$c^2 \circ a^2 + b^2$$

$$29^2 \circ 20^2 + 21^2$$

$$841 \circ 400 + 441$$

$$841 \equiv 841$$

Right



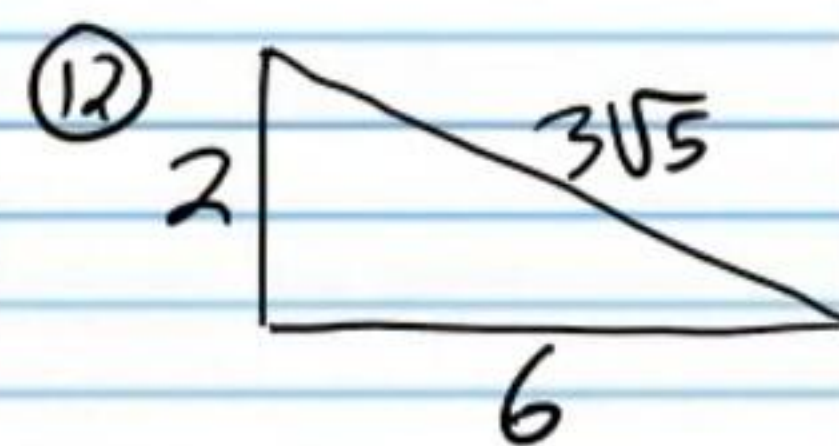
$$c^2 \circ a^2 + b^2$$

$$14^2 \circ 12^2 + 8^2$$

$$196 \circ 144 + 64$$

$$196 < 208$$

Acute



$$c^2 \circ a^2 + b^2$$

$$(3\sqrt{5})^2 \circ 2^2 + 6^2$$

$$45 \circ 4 + 36$$

$$45 > 40$$

Obtuse

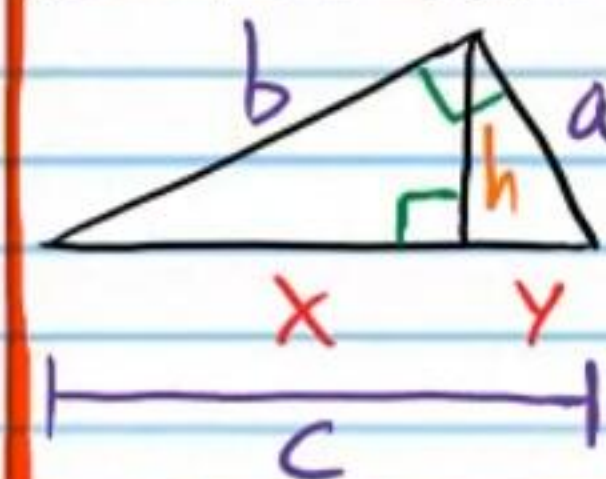
Geometric Mean Theorems

Geometric Mean: The n th root of the product of n numbers.

Example I) 3, 4, 12 Geometric Mean = $\sqrt[3]{(3)(4)(12)} \approx 5.241$

Example II) 10, 7 Geometric Mean = $\sqrt{(10)(7)} \approx 8.3667$

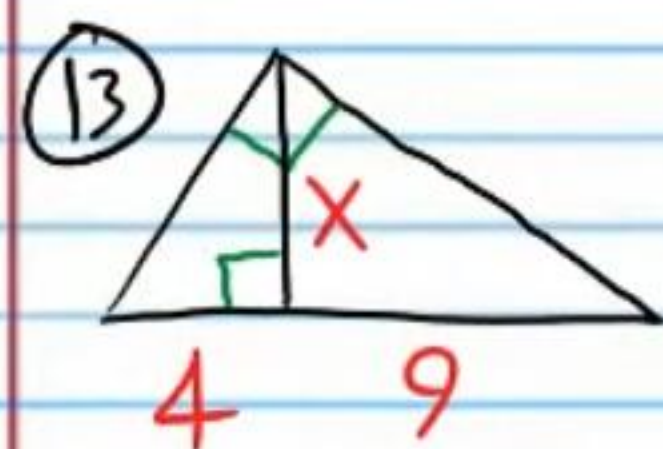
Geometric Mean Theorem for Altitudes: The altitude of a right triangle extending from the vertex of the right angle to the hypotenuse is the geometric mean of the two segments on the hypotenuse created by contact with the altitude.



$$h = \sqrt{xy}$$

Geometric Mean Theorem for Legs: $a = \sqrt{cy}$, $b = \sqrt{cx}$

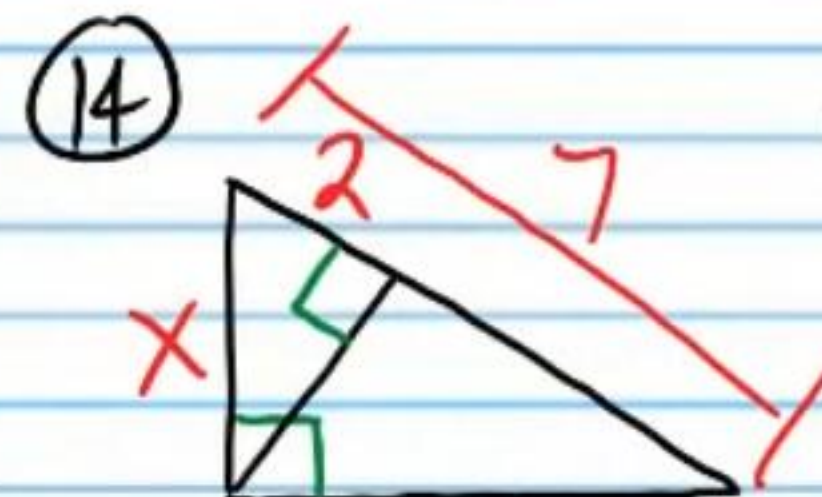
Find the unknown values.



$$x = \sqrt{4 \cdot 9}$$

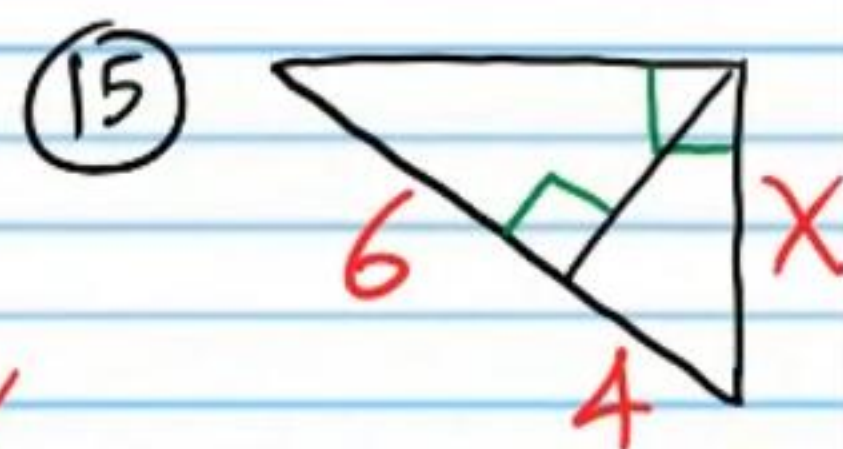
$$x = \sqrt{36}$$

$$\boxed{x = 6}$$



$$x = \sqrt{2 \cdot 7}$$

$$\boxed{x = \sqrt{14}}$$



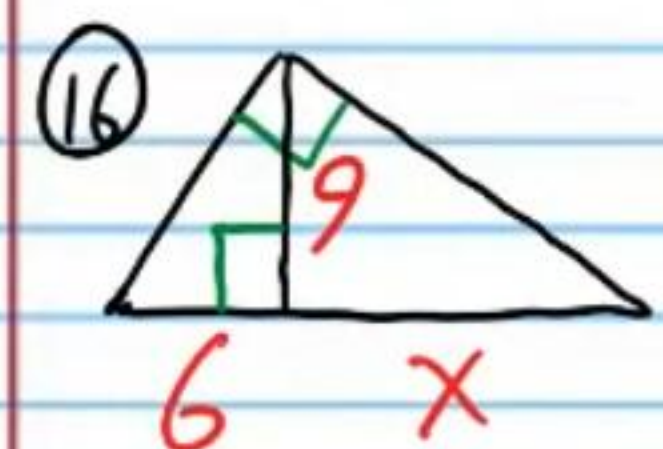
$$x = \sqrt{4(6+4)}$$

$$x = \sqrt{4(10)}$$

$$x = \sqrt{4 \cdot 10}$$

$$x = 2 \cdot \sqrt{10}$$

$$\boxed{x = 2\sqrt{10}}$$



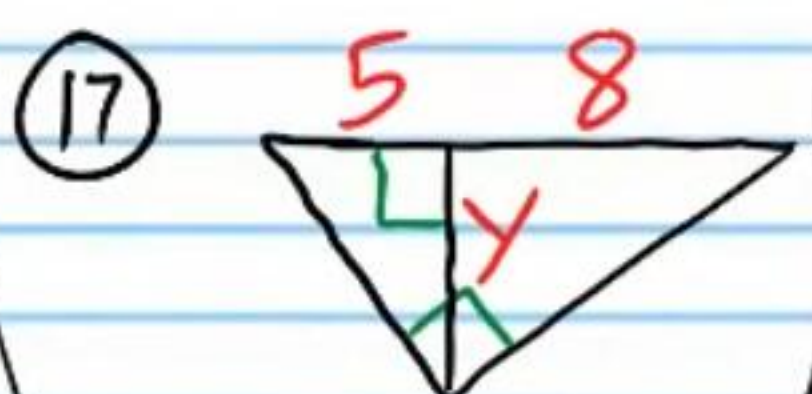
$$9 = \sqrt{6x}$$

$$9^2 = (\sqrt{6x})^2$$

$$81 = 6x$$

$$\frac{81}{6} = x$$

$$\boxed{\frac{27}{2} = x}$$



$$y = \sqrt{5 \cdot 8}$$

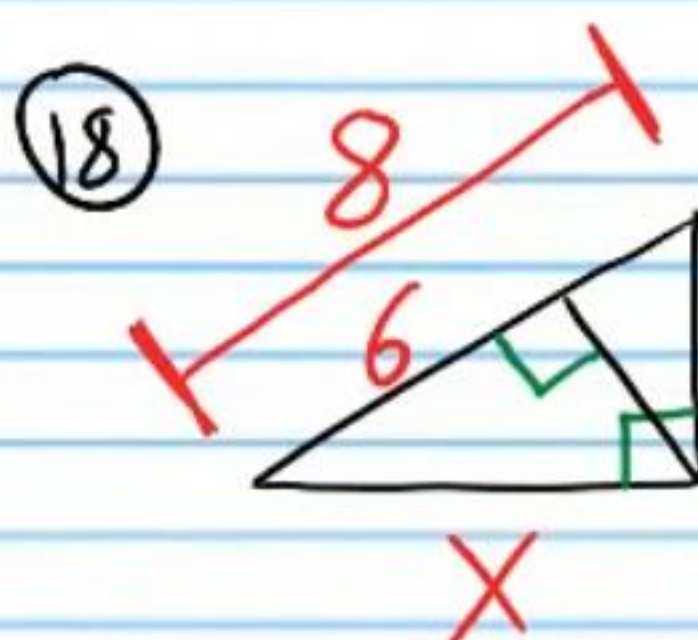
$$y = \sqrt{40}$$

$$y = \sqrt{4 \cdot 10}$$

$$y = \sqrt{4 \cdot 10}$$

$$y = 2 \cdot \sqrt{10}$$

$$\boxed{y = 2\sqrt{10}}$$

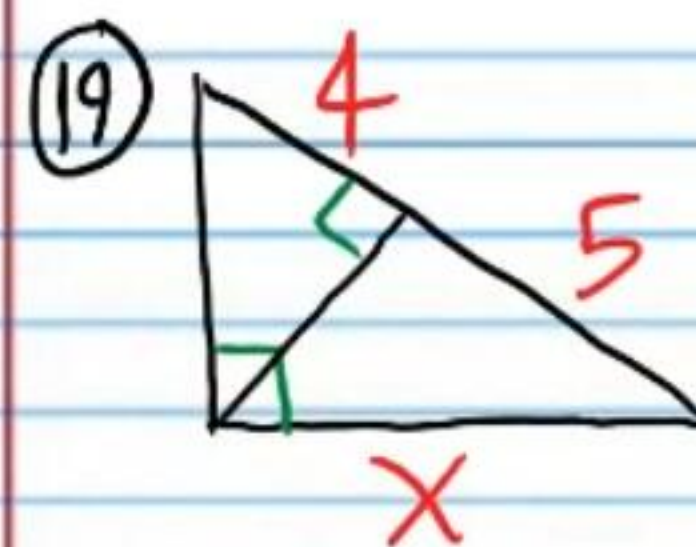


$$x = \sqrt{6 \cdot 8}$$

$$x = \sqrt{48}$$

$$x = \sqrt{16 \cdot 3}$$

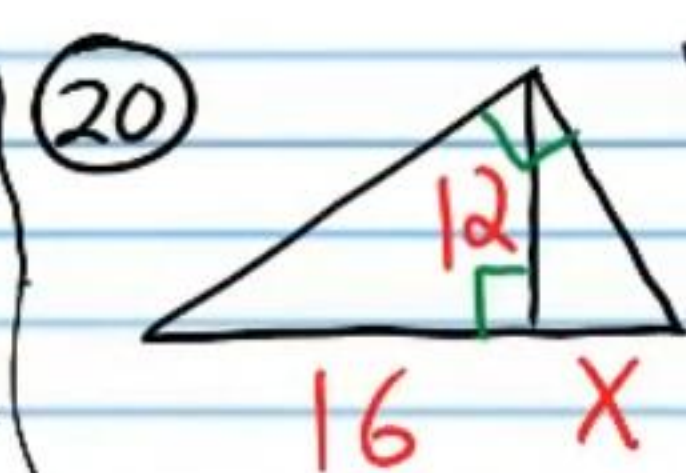
$$\boxed{x = 4\sqrt{3}}$$



$$x = \sqrt{5(4+5)}$$

$$x = \sqrt{5(9)}$$

$$\boxed{x = 3\sqrt{5}}$$

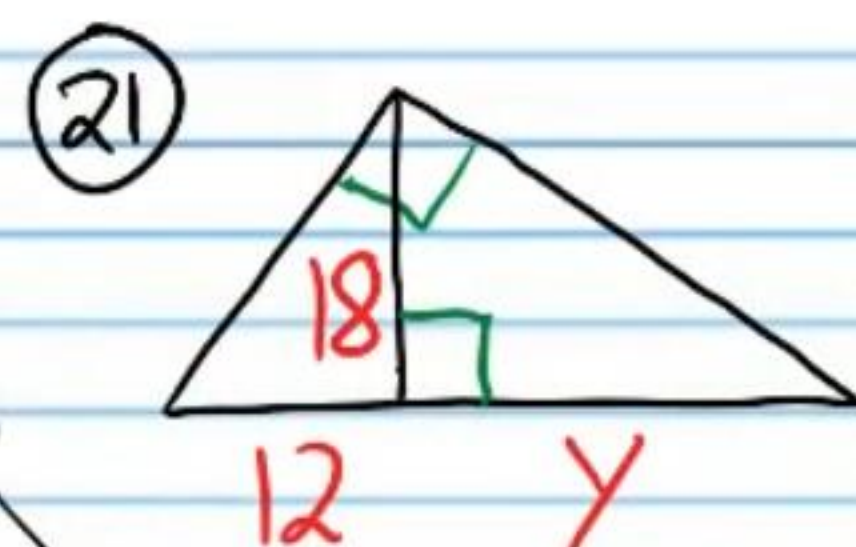


$$12 = \sqrt{16x}$$

$$12^2 = (\sqrt{16x})^2$$

$$144 = 16x$$

$$\boxed{9 = x}$$



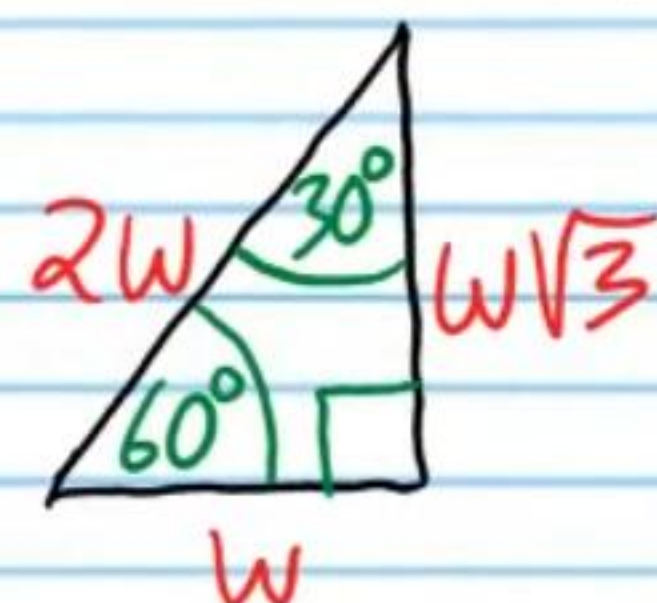
$$18 = \sqrt{12y}$$

$$324 = 12y$$

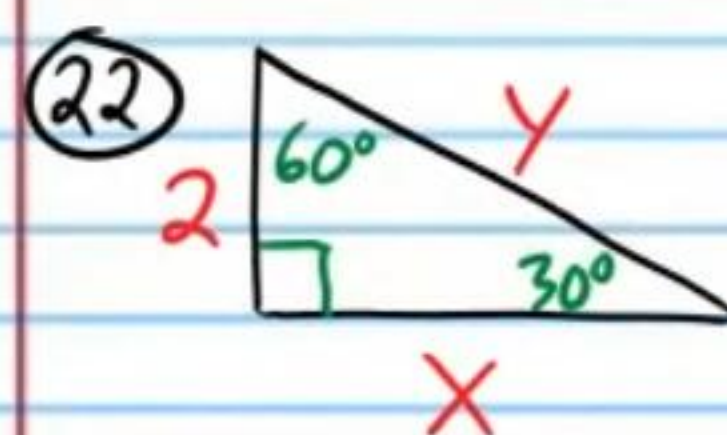
$$\boxed{27 = y}$$

Special Right Triangles

30°-60°-90° Triangle Theorem: If the inner angles of a triangle are 30°, 60°, and 90°, and the length of the side opposite the 30° angle is w , then the length of the hypotenuse is $2w$ and the length of the side opposite the 60° angle is $w\sqrt{3}$.

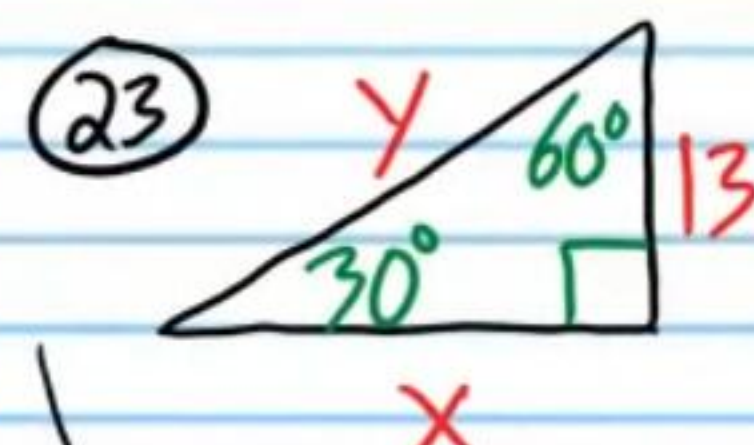


Find the unknown values.



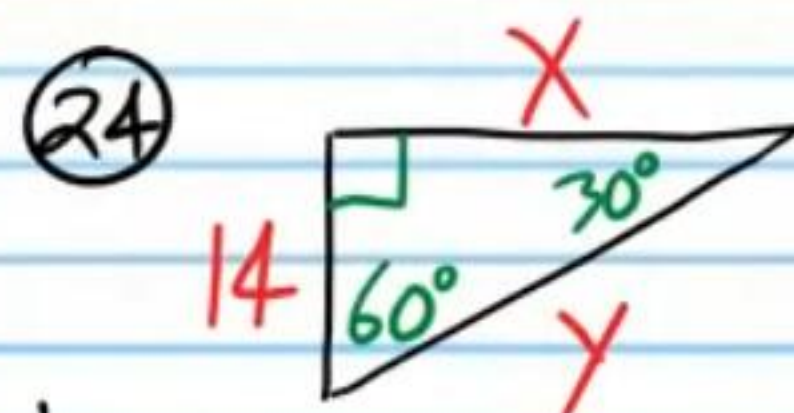
$$y = 2(2) = 4$$

$$x = 2\sqrt{3}$$



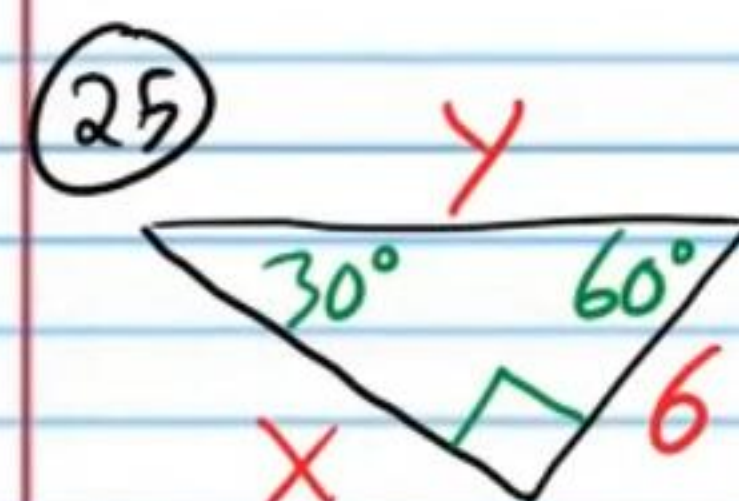
$$y = 2(13) = 26$$

$$x = 13\sqrt{3}$$



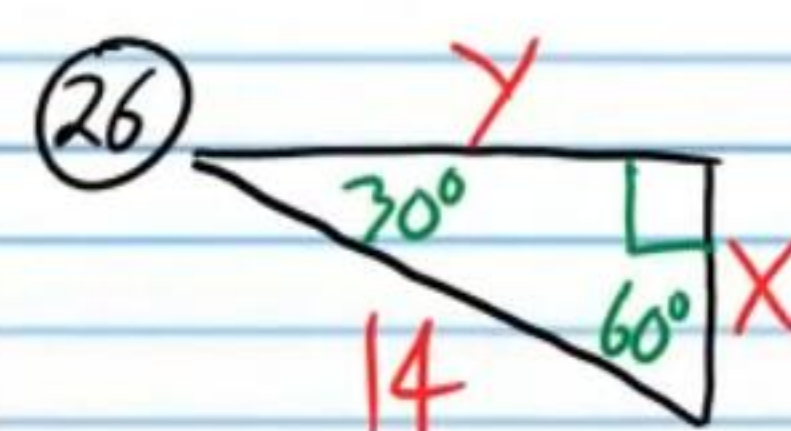
$$y = 2(14) = 28$$

$$x = 14\sqrt{3}$$



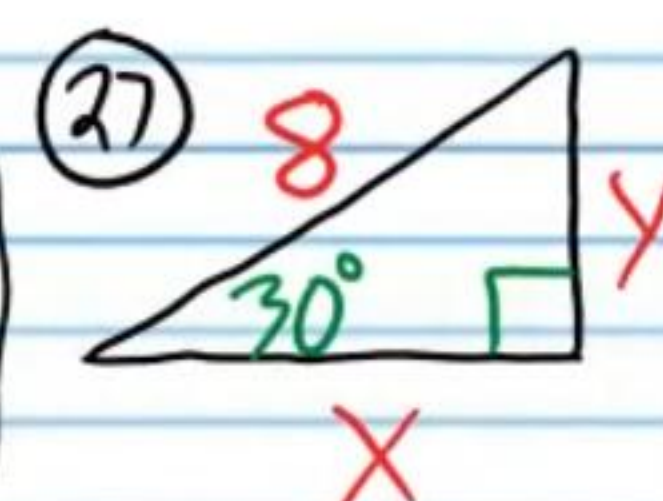
$$y = 2(6) = 12$$

$$x = 6\sqrt{3}$$



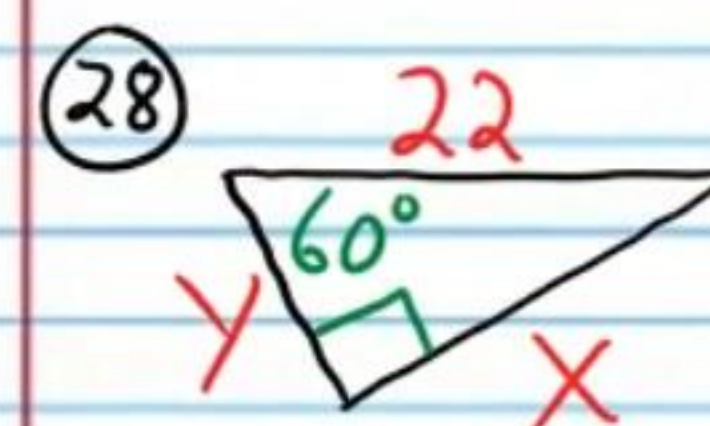
$$x = 14/2 = 7$$

$$y = 7\sqrt{3}$$



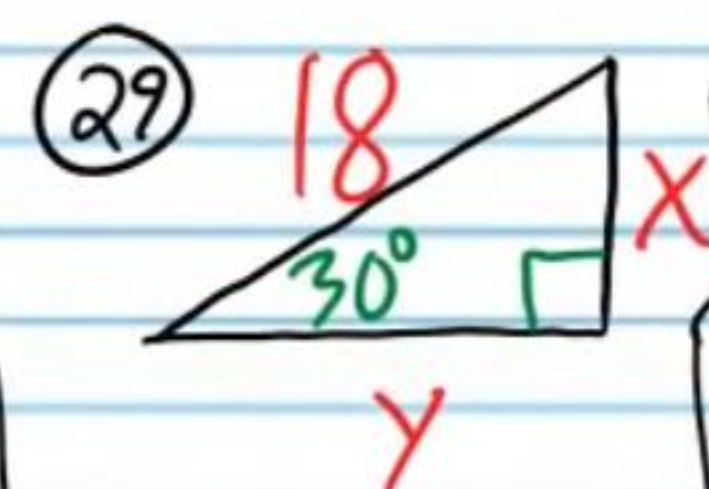
$$y = 8/2 = 4$$

$$x = 4\sqrt{3}$$



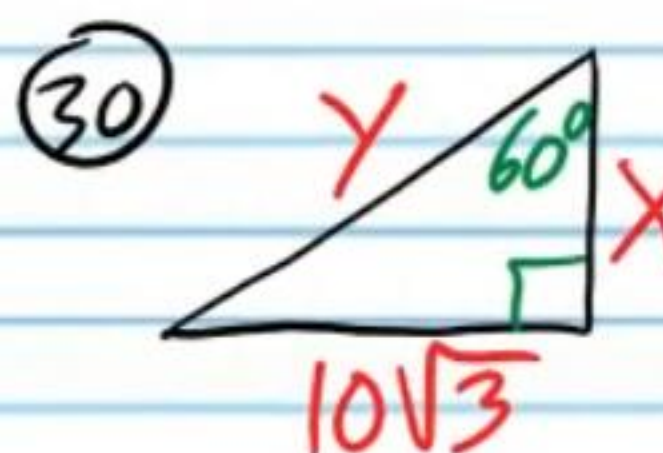
$$y = 22/2 = 11$$

$$x = 11\sqrt{3}$$



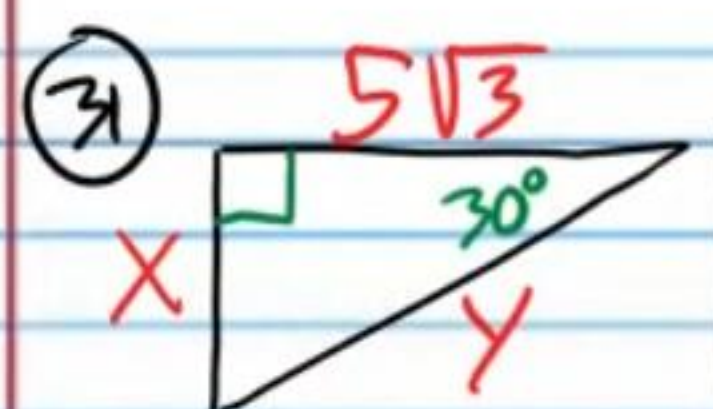
$$x = \frac{18}{2} = 9$$

$$y = 9\sqrt{3}$$



$$x = 10$$

$$y = 10(2) = 20$$



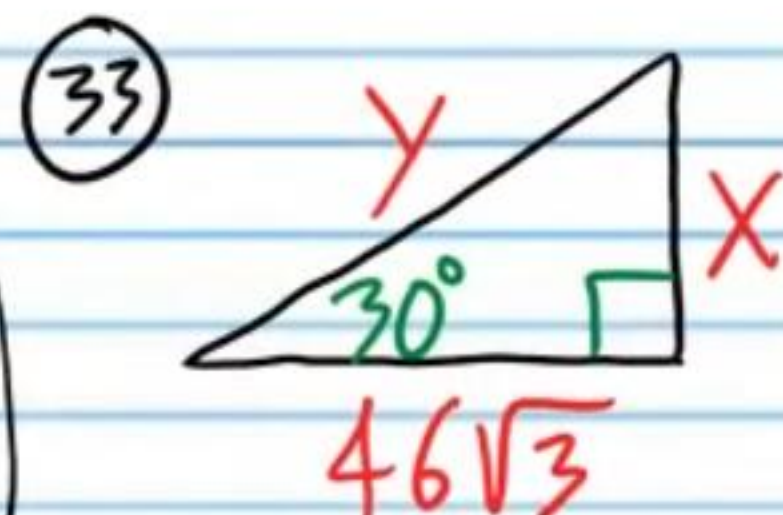
$$X = 5$$

$$Y = 5(2) = 10$$



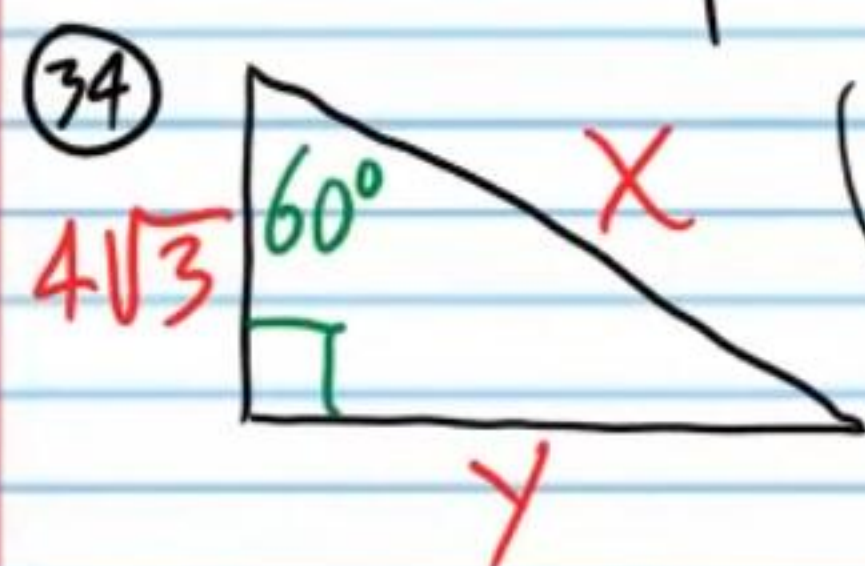
$$X = 11$$

$$Y = 11(2) = 22$$



$$X = 46$$

$$Y = 2(46) = 92$$



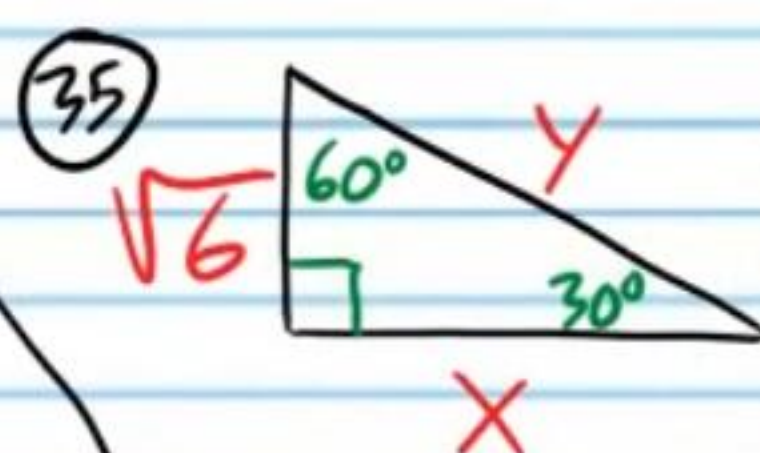
$$X = 2(4\sqrt{3}) = 8\sqrt{3}$$

$$Y = (4\sqrt{3})(\sqrt{3})$$

$$Y = 4(\sqrt{3})^2$$

$$Y = 4 \cdot 3$$

$$Y = 12$$



$$Y = 2\sqrt{6}$$

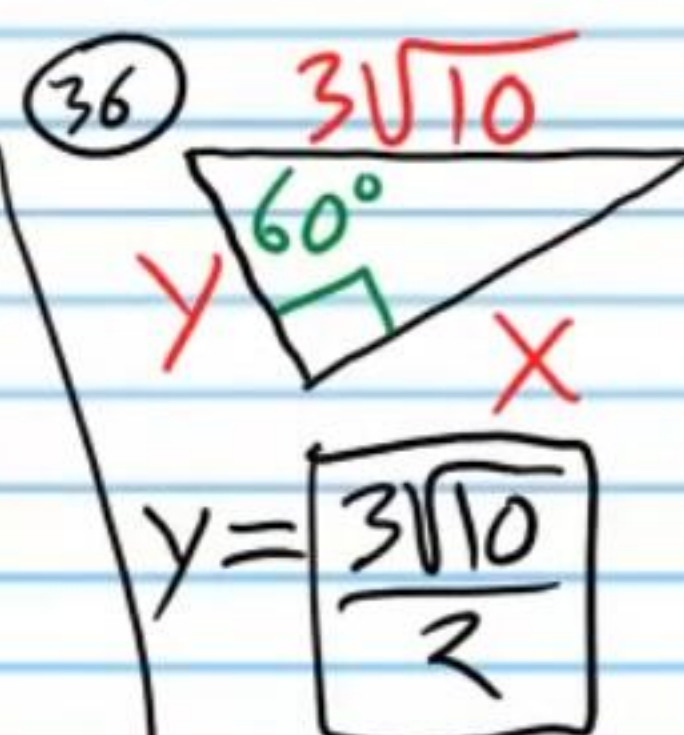
$$X = (\sqrt{6})(\sqrt{3})$$

$$X = \sqrt{18}$$

$$X = \sqrt{9 \cdot 2}$$

$$X = \sqrt{9} \cdot \sqrt{2}$$

$$X = 3\sqrt{2}$$

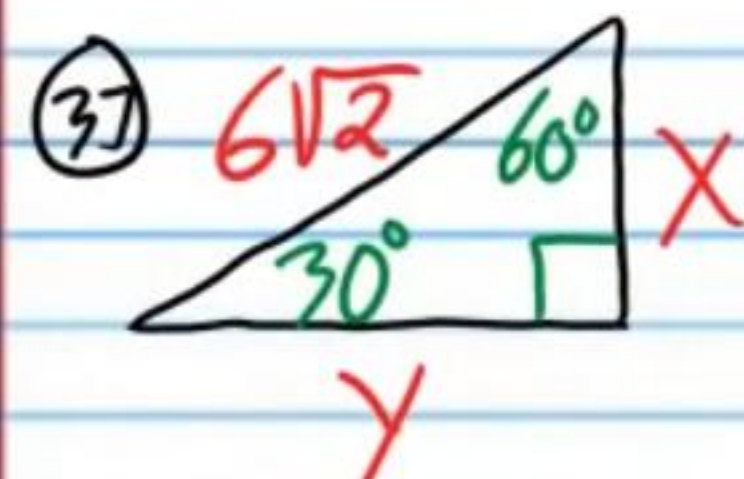


$$Y = \frac{3\sqrt{10}}{2}$$

$$X = \frac{3\sqrt{10}}{2}(\sqrt{3})$$

$$X = \frac{3\sqrt{10} \cdot \sqrt{3}}{2}$$

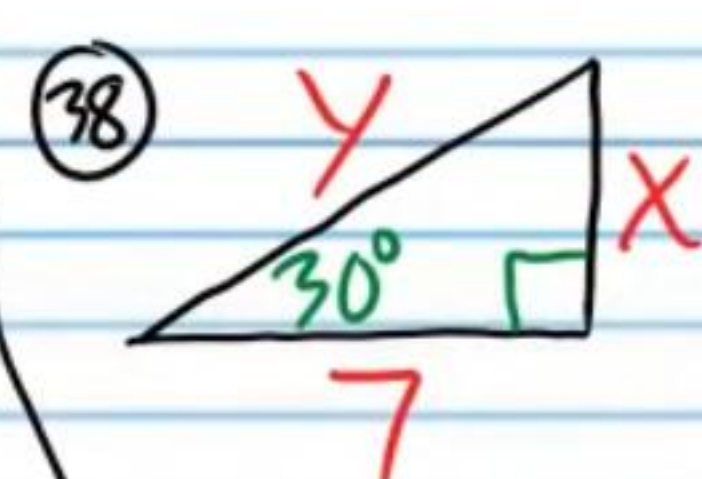
$$X = \frac{3\sqrt{30}}{2}$$



$$X = \frac{6\sqrt{2}}{2} = 3\sqrt{2}$$

$$Y = 3\sqrt{2}(\sqrt{3})$$

$$Y = 3\sqrt{6}$$



$$7 = X\sqrt{3}$$

$$\frac{7}{\sqrt{3}} = X$$

$$\frac{\sqrt{3} \cdot 7}{\sqrt{3} \cdot \sqrt{3}} = X$$

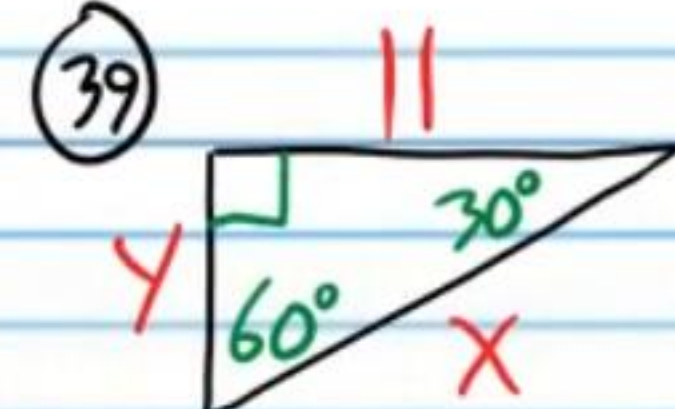
$$\frac{7\sqrt{3}}{\sqrt{9}} = X$$

$$\frac{7\sqrt{3}}{3} = X$$

$$Y = 2\left(\frac{7\sqrt{3}}{3}\right)$$

$$Y = \frac{2}{1}\left(\frac{7\sqrt{3}}{3}\right)$$

$$Y = \frac{14\sqrt{3}}{3}$$



$$11 = Y\sqrt{3}$$

$$\frac{11}{\sqrt{3}} = Y$$

$$\frac{\sqrt{3} \cdot 11}{\sqrt{3} \cdot \sqrt{3}} = Y$$

$$\frac{11\sqrt{3}}{\sqrt{9}} = Y$$

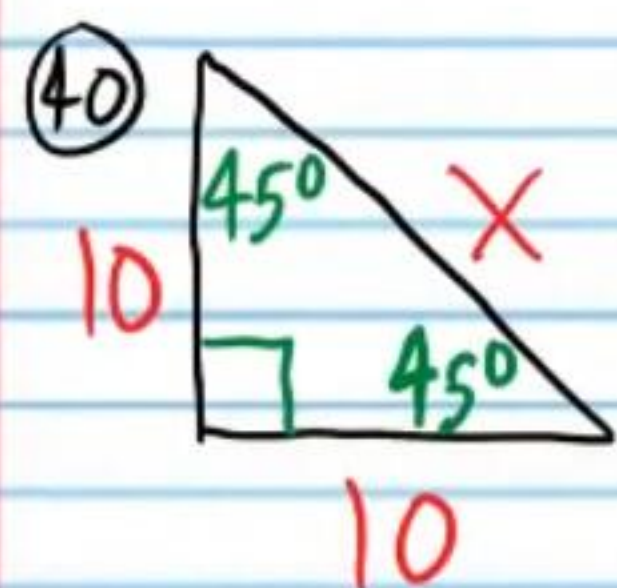
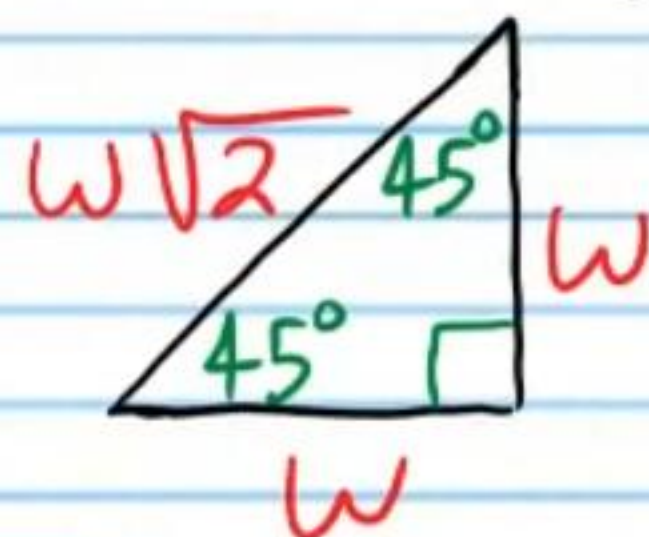
$$\frac{11\sqrt{3}}{3} = Y$$

$$X = \frac{11\sqrt{3}}{3}(2)$$

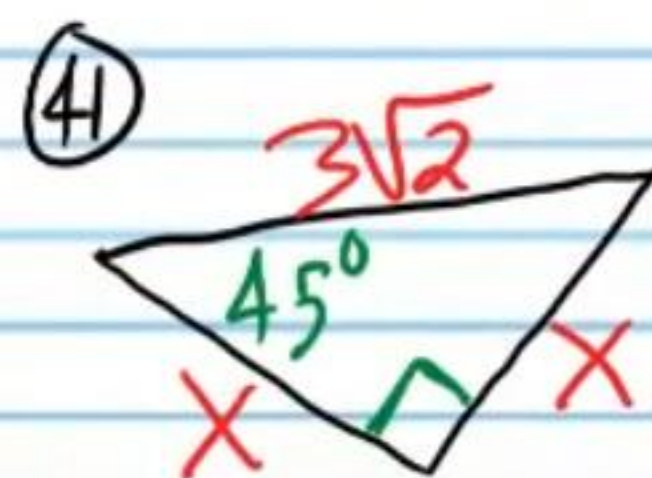
$$X = \frac{11\sqrt{3}}{3}\left(\frac{2}{1}\right)$$

$$X = \frac{22\sqrt{3}}{3}$$

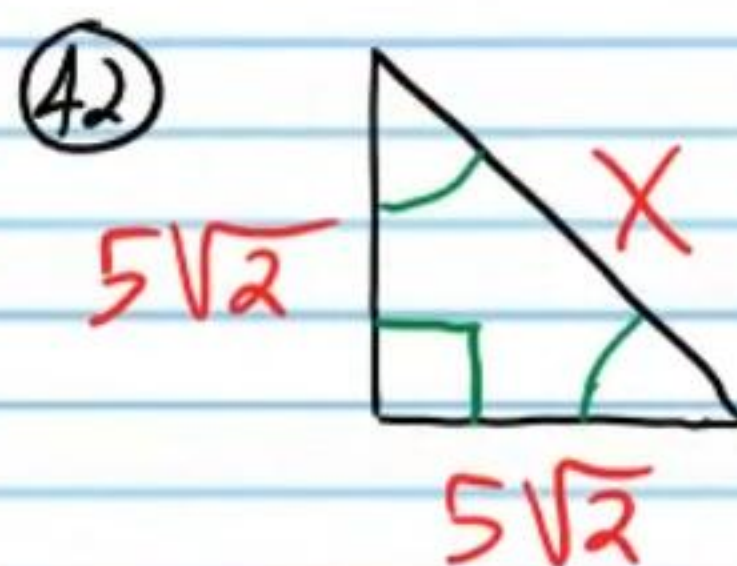
45°-45°-90° Triangle Theorem: If the inner angles of a triangle are 45°, 45°, and 90°, and the length of the two sides opposite the 45° angles are w , then the length of the hypotenuse is $w\sqrt{2}$.



$$X = 10\sqrt{2}$$



$$X = 3$$

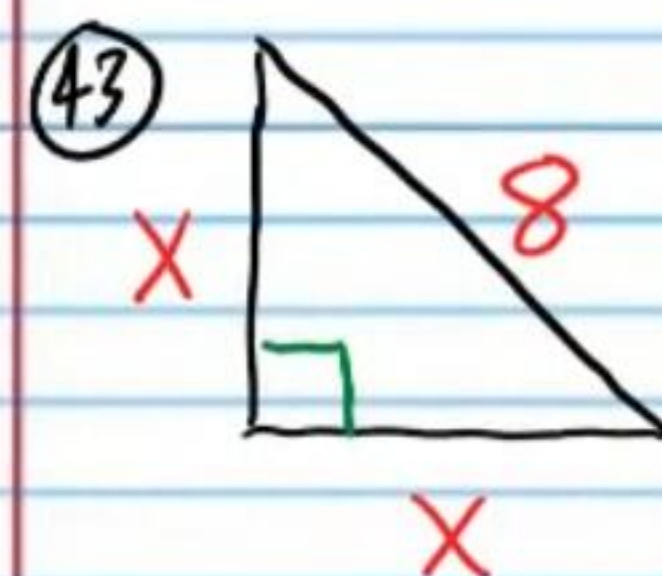


$$X = (5\sqrt{2})(\sqrt{2})$$

$$X = 5\sqrt{4}$$

$$X = 5 \cdot 2$$

$$X = 10$$



$$8 = X\sqrt{2}$$

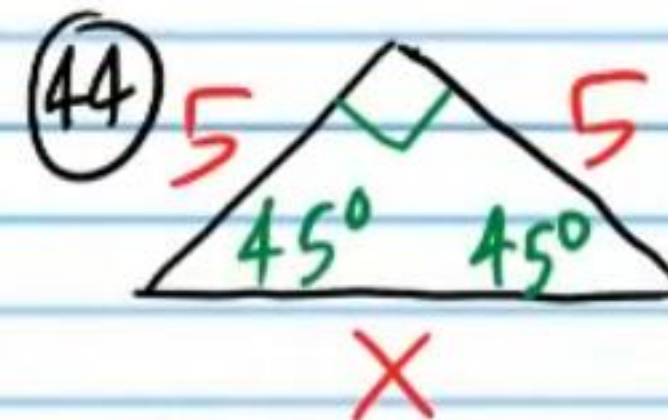
$$\frac{8}{\sqrt{2}} = X$$

$$\frac{\sqrt{2} \cdot 8}{\sqrt{2} \cdot \sqrt{2}} = X$$

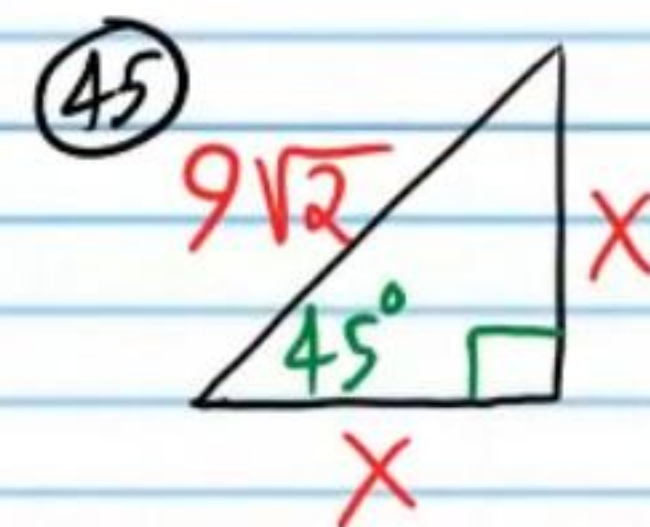
$$\frac{8\sqrt{2}}{\sqrt{4}} = X$$

$$\frac{8\sqrt{2}}{2} = X$$

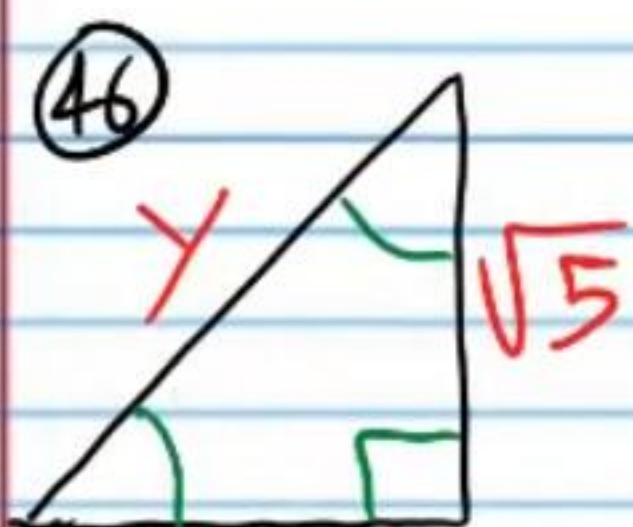
$$4\sqrt{2} = X$$



$$X = 5\sqrt{2}$$



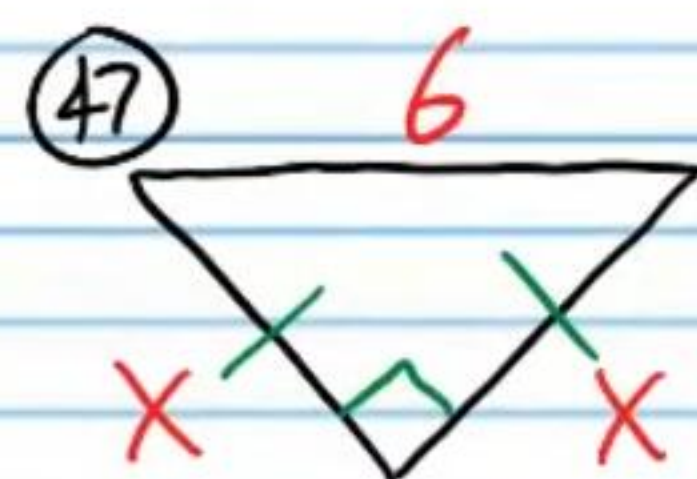
$$X = 9$$



$$x = \boxed{\sqrt{5}}$$

$$y = (\sqrt{5})(\sqrt{2})$$

$$y = \boxed{\sqrt{10}}$$



$$6 = x\sqrt{2}$$

$$\frac{6}{\sqrt{2}} = x$$

$$\frac{\sqrt{2} \cdot 6}{\sqrt{2} \cdot \sqrt{2}} = x$$

$$\frac{6\sqrt{2}}{\sqrt{4}} = x$$

$$\frac{6\sqrt{2}}{2} = x$$

$$\boxed{3\sqrt{2}} = x$$